In[159]:=

```
Clear["`*"]
```

In[160]:=

```
<< NumericalCalculus`
KerrRad[a_, p_, e_, i_, ν_, Θ_, ϕ_, M_] := {
   (*First, define the orbital energy E, angular momentum L,
   and Carter constant Q in terms of the intrinsic binary parameters (a, e, p, i).*)
   (*Below follows Appendix B of Schmidt: gr-qc/0202090 *)
   (zm = (Cos[(Pi / 2) - (i * (Pi / 180))])^2);
   (*Note: This corresponds to z_^2 in Schmidt*)
   (ra = p / (1 - e));
   (rp = p / (1 + e));
   (Δ = r^2 - 2 r + a^2);
   (*Defining coefficients to colve for orbital constants E, L, Q*)
   (f = r^4 + a^2 * (r * (r + 2) + Δ * zm)); (g = 2 * a * r);
   (h = r * (r - 2) + Δ * (zm) / (1 - zm));
   (d = (r^2 + a^2 * zm) * Δ);
   ((d1 = d /. r → rp);
    (d2 = d /. r → ra);
    (f1 = f /. r → rp);
    (f2 = f /. r → ra);
    (g1 = g /. r → rp);
    (g2 = g /. r → ra);
    (h1 = h /. r → rp);
    (h2 = h /. r → ra));
   (k = Det[{{d1, h1}, {d2, h2}}]) (rho = Det[{{f1, h1}, {f2, h2}}]);
   (sig = Det[{{g1, h1}, {g2, h2}}]);
   (ep = Det[{{d1, g1}, {d2, g2}}]);
   (nu = Det[{{f1, g1}, {f2, g2}}]);
   (EE = En /.
      Solve[(rho^2 + 4 * nu * sig) * En^4 - 2 * (k * rho + 2 * ep * sig) * En^2 + k^2 == 0, En]);
   (*The roots of the above orbital energy are sorted
    to determine the prograde and retrograde orbital parameters.*)

   (*Of the four roots: -E^(p), -E^(r), E^(p),
   E^(r) (where p and are refer to prograde and retrograde), E^(p) is the lower,
   positive energy value. Thus, Epro is assigned to the third sorted value.*)
   (EE2 = Sort[EE]); (Epro = EE2[[3]]);

   (*The prograde angular momentum is the higher of the two values once the
    prograde energy is substituted into the equation. Q is computed directly.*)
   (P = Lz /. Solve[(-h1) * Lz^2 - (2 * g1 * Epro) * Lz + (f1 * Epro^2) - d1 == 0, Lz]);
   (P2 = Sort[P]);
   (Lpro = P2[[2]]);
   ((Q = ( (1 - Epro^2) * a^2 + Lpro^2 / (1 - zm)) * zm));
   S = a;
```

```
(*Need to test which of the roots from the radial potential are the apastron,
Ra, and periastron, Rp*)
E0 = Epro; L0 = Lpro; q0 = Q;
Root10 =
 Re[-1/(2 (-1 + E0^2)) +
  1/2 Sqrt[1/(-1 + E0^2)^2 - (2 (-S^2 + S^2 E0^2 - L0^2 - q0))/(3 (-1 + E0^2)) +
    (2^(1/3) ((-S^2 + S^2 E0^2 - L0^2 - q0)^2 - 12
          S^2 (-1 + E0^2) q0 - 12 (S^2 E0^2 - 2 S E0 L0 + L0^2 + q0)))/
     (3 (-1 + E0^2) (2 (-S^2 + S^2 E0^2 - L0^2 - q0)^3 - 108 S^2 q0 + 72 S^2 (-1 + E0^2) (-S^2 + S^2 E0^2 - L0^2 - q0) q0 -
        36 (-S^2 + S^2 E0^2 - L0^2 - q0) (S^2 E0^2 - 2 S E0 L0 + L0^2 + q0) +
        108 (-1 + E0^2) (S^2 E0^2 - 2 S E0 L0 + L0^2 + q0)^2 + Sqrt[-4 ((-S^2 + S^2 E0^2 - L0^2 - q0)^2 -
            12 S^2 (-1 + E0^2) q0 - 12 (S^2 E0^2 - 2 S E0 L0 + L0^2 + q0))^3 +
          (2 (-S^2 + S^2 E0^2 - L0^2 - q0)^3 - 108 S^2 q0 + 72 S^2 (-1 + E0^2) (-S^2 +
              S^2 E0^2 - L0^2 - q0) q0 - 36 (-S^2 + S^2 E0^2 - L0^2 - q0) (S^2 E0^2 - 2 S E0 L0 +
              L0^2 + q0) + 108 (-1 + E0^2) (S^2 E0^2 - 2 S E0 L0 + L0^2 + q0)^2)^2])^(1/3)) +
    1/(3 × 2^(1/3) (-1 + E0^2)) (2 (-S^2 + S^2 E0^2 - L0^2 - q0)^3 - 108 S^2 q0 + 72 S^2 (-1 + E0^2)
       (-S^2 + S^2 E0^2 - L0^2 - q0) q0 - 36 (-S^2 + S^2 E0^2 - L0^2 - q0)
       (S^2 E0^2 - 2 S E0 L0 + L0^2 + q0) + 108 (-1 + E0^2) (S^2 E0^2 - 2 S E0 L0 + L0^2 + q0)^2 +
      Sqrt[-4 ((-S^2 + S^2 E0^2 - L0^2 - q0)^2 - 12 S^2 (-1 + E0^2) q0 - 12 (S^2 E0^2 - 2 S E0 L0 +
            L0^2 + q0))^3 + (2 (-S^2 + S^2 E0^2 - L0^2 - q0)^3 - 108 S^2 q0 + 72 S^2 (-1 + E0^2)
          (-S^2 + S^2 E0^2 - L0^2 - q0) q0 - 36 (-S^2 + S^2 E0^2 - L0^2 - q0) (S^2 E0^2 - 2 S E0
            L0 + L0^2 + q0) + 108 (-1 + E0^2) (S^2 E0^2 - 2 S E0 L0 + L0^2 + q0)^2)^2])^(1/3)) +
  1/2 Sqrt[2/(-1 + E0^2)^2 - (4 (-S^2 + S^2 E0^2 - L0^2 - q0))/(3 (-1 + E0^2)) - (2^(1/3) ((-S^2 + S^2 E0^2 - L0^2 - q0)^2 -
        12 S^2 (-1 + E0^2) q0 - 12 (S^2 E0^2 - 2 S E0 L0 + L0^2 + q0)))/
     (3 (-1 + E0^2) (2 (-S^2 + S^2 E0^2 - L0^2 - q0)^3 - 108 S^2 q0 + 72 S^2 (-1 + E0^2)
         (-S^2 + S^2 E0^2 - L0^2 - q0) q0 - 36 (-S^2 + S^2 E0^2 - L0^2 - q0)
         (S^2 E0^2 - 2 S E0 L0 + L0^2 + q0) + 108 (-1 + E0^2) (S^2 E0^2 - 2 S E0 L0 + L0^2 + q0)^2 +
```

$$\sqrt{\left(-4\left(\left(-S^2+S^2\,E0^2-L0^2-q0\right)^2-12\,S^2\left(-1+E0^2\right)q0-12\left(S^2\,E0^2-2\,S\,E0\,L0+L0^2+q0\right)\right)^3+\left(2\left(-S^2+S^2\,E0^2-L0^2-q0\right)^3-108\,S^2\,q0+72\,S^2\left(-1+E0^2\right)\left(-S^2+S^2\,E0^2-L0^2-q0\right)q0-36\left(-S^2+S^2\,E0^2-L0^2-q0\right)\left(S^2\,E0^2-2\,S\,E0\,L0+L0^2+q0\right)+108\left(-1+E0^2\right)\left(S^2\,E0^2-2\,S\,E0\,L0+L0^2+q0\right)^2\right)^2\right)}\Bigg)^{1/3}\Bigg)-\frac{1}{3\times 2^{1/3}\left(-1+E0^2\right)}\left(2\left(-S^2+S^2\,E0^2-L0^2-q0\right)^3-108\,S^2\,q0+72\,S^2\left(-1+E0^2\right)\left(-S^2+S^2\,E0^2-L0^2-q0\right)q0-36\left(-S^2+S^2\,E0^2-L0^2-q0\right)\left(S^2\,E0^2-2\,S\,E0\,L0+L0^2+q0\right)+108\left(-1+E0^2\right)\left(S^2\,E0^2-2\,S\,E0\,L0+L0^2+q0\right)^2+\sqrt{\left(-4\left(\left(-S^2+S^2\,E0^2-L0^2-q0\right)^2-12\,S^2\left(-1+E0^2\right)q0-12\left(S^2\,E0^2-2\,S\,E0\,L0+L0^2+q0\right)\right)^3+\left(2\left(-S^2+S^2\,E0^2-L0^2-q0\right)^3-108\,S^2\,q0+72\,S^2\left(-1+E0^2\right)\left(-S^2+S^2\,E0^2-L0^2-q0\right)q0-36\left(-S^2+S^2\,E0^2-L0^2-q0\right)\left(S^2\,E0^2-2\,S\,E0\,L0+L0^2+q0\right)+108\left(-1+E0^2\right)\left(S^2\,E0^2-2\,S\,E0\,L0+L0^2+q0\right)^2\right)^2\right)}\Bigg)^{1/3}+\left(-\frac{8}{\left(-1+E0^2\right)^3}+\frac{8\left(-S^2+S^2\,E0^2-L0^2-q0\right)}{\left(-1+E0^2\right)^2}-\frac{16\left(S^2\,E0^2-2\,S\,E0\,L0+L0^2+q0\right)}{-1+E0^2}\right)\Bigg/\left(4\sqrt{\left(\frac{1}{\left(-1+E0^2\right)^2}-\frac{2\left(-S^2+S^2\,E0^2-L0^2-q0\right)}{3\left(-1+E0^2\right)}+\left(2^{1/3}\left(\left(-S^2+S^2\,E0^2-L0^2-q0\right)^2-12\,S^2\left(-1+E0^2\right)q0-12\left(S^2\,E0^2-2\,S\,E0\,L0+L0^2+q0\right)\right)\Bigg/\left(3\left(-1+E0^2\right)\left(2\left(-S^2+S^2\,E0^2-L0^2-q0\right)^3-108\,S^2\,q0+72\,S^2\left(-1+E0^2\right)\left(-S^2+S^2\,E0^2-L0^2-q0\right)q0-36\left(-S^2+S^2\,E0^2-L0^2-q0\right)\left(S^2\,E0^2-2\,S\,E0\,L0+L0^2+q0\right)+108\left(-1+E0^2\right)\left(S^2\,E0^2-2\,S\,E0\,L0+L0^2+q0\right)^2+\sqrt{\left(-4\left(\left(-S^2+S^2\,E0^2-L0^2-q0\right)^2-12\,S^2\left(-1+E0^2\right)q0-12\left(S^2\,E0^2-2\,S\,E0\,L0+L0^2+q0\right)\right)^3+\left(2\left(-S^2+S^2\,E0^2-L0^2-q0\right)^3-108\,S^2\,q0+72\,S^2\left(-1+E0^2\right)\left(-S^2+S^2\,E0^2-L0^2-q0\right)q0-36\left(-S^2+S^2\,E0^2-L0^2-q0\right)\left(S^2\,E0^2-2\,S\,E0\,L0+L0^2+q0\right)+108\left(-1+E0^2\right)\left(S^2\,E0^2-2\,S\,E0\,L0+L0^2+q0\right)^2\right)^2\right)}\Bigg)^{1/3}\Bigg)+\frac{1}{3\times 2^{1/3}\left(-1+E0^2\right)}\left(2\left(-S^2+S^2\,E0^2-L0^2-q0\right)^3-108\,S^2\,q0+72\,S^2\left(-1+E0^2\right)\left(-S^2+S^2\,E0^2-L0^2-q0\right)q0-36\left(-S^2+S^2\,E0^2-L0^2-q0\right)\left(S^2\,E0^2-2\,S\,E0\,L0+L0^2+q0\right)+108\left(-1+E0^2\right)\left(S^2\,E0^2-2\,S\,E0\,L0+L0^2+q0\right)^2+\sqrt{\left(-4\left(\left(-S^2+S^2\,E0^2-L0^2-q0\right)^2-12\,S^2\left(-1+E0^2\right)q0-\right.\right.\right.$$

$$12\left(S^2\,E0^2-2\,S\,E0\,L0+L0^2+q0\right)\Big)^3+\Big(2\left(-S^2+S^2\,E0^2-L0^2-q0\right)^3-108\,S^2\,q0+72\,S^2\left(-1+E0^2\right)\left(-S^2+S^2\,E0^2-L0^2-q0\right)q0-36\left(-S^2+S^2\,E0^2-L0^2-q0\right)\left(S^2\,E0^2-2\,S\,E0\,L0+L0^2+q0\right)+108\left(-1+E0^2\right)\left(S^2\,E0^2-2\,S\,E0\,L0+L0^2+q0\right)^2\Big)^2\Big)^{1/3}\Big)\Big)\Big)\Big];$$

Root20 =

$$\mathrm{Re}\Big[-\frac{1}{2\left(-1+E0^2\right)}-\frac{1}{2}\sqrt{\Big(\frac{1}{\left(-1+E0^2\right)^2}-\frac{2\left(-S^2+S^2\,E0^2-L0^2-q0\right)}{3\left(-1+E0^2\right)}+\Big(2^{1/3}\Big(\left(-S^2+S^2\,E0^2-L0^2-q0\right)^2-12\,S^2\left(-1+E0^2\right)q0-12\left(S^2\,E0^2-2\,S\,E0\,L0+L0^2+q0\right)\Big)\Big/\Big(3\left(-1+E0^2\right)\Big(2\left(-S^2+S^2\,E0^2-L0^2-q0\right)^3-108\,S^2\,q0+72\,S^2\left(-1+E0^2\right)\left(-S^2+S^2\,E0^2-L0^2-q0\right)q0-36\left(-S^2+S^2\,E0^2-L0^2-q0\right)\left(S^2\,E0^2-2\,S\,E0\,L0+L0^2+q0\right)+108\left(-1+E0^2\right)\left(S^2\,E0^2-2\,S\,E0\,L0+L0^2+q0\right)^2+\sqrt{\Big(-4\Big(\left(-S^2+S^2\,E0^2-L0^2-q0\right)^2-12\,S^2\left(-1+E0^2\right)q0-12\left(S^2\,E0^2-2\,S\,E0\,L0+L0^2+q0\right)\Big)^3+\Big(2\left(-S^2+S^2\,E0^2-L0^2-q0\right)^3-108\,S^2\,q0+72\,S^2\left(-1+E0^2\right)\left(-S^2+S^2\,E0^2-L0^2-q0\right)q0-36\left(-S^2+S^2\,E0^2-L0^2-q0\right)\left(S^2\,E0^2-2\,S\,E0\,L0+L0^2+q0\right)+108\left(-1+E0^2\right)\left(S^2\,E0^2-2\,S\,E0\,L0+L0^2+q0\right)^2\Big)^2\Big)}\Big)^{1/3}\Big)\Big)+\frac{1}{3\times 2^{1/3}\left(-1+E0^2\right)}\Big(2\left(-S^2+S^2\,E0^2-L0^2-q0\right)^3-108\,S^2\,q0+72\,S^2\left(-1+E0^2\right)\left(-S^2+S^2\,E0^2-L0^2-q0\right)q0-36\left(-S^2+S^2\,E0^2-L0^2-q0\right)\left(S^2\,E0^2-2\,S\,E0\,L0+L0^2+q0\right)+108\left(-1+E0^2\right)\left(S^2\,E0^2-2\,S\,E0\,L0+L0^2+q0\right)^2+\sqrt{\Big(-4\Big(\left(-S^2+S^2\,E0^2-L0^2-q0\right)^2-12\,S^2\left(-1+E0^2\right)q0-12\left(S^2\,E0^2-2\,S\,E0\,L0+L0^2+q0\right)\Big)^3+\Big(2\left(-S^2+S^2\,E0^2-L0^2-q0\right)^3-108\,S^2\,q0+72\,S^2\left(-1+E0^2\right)\left(-S^2+S^2\,E0^2-L0^2-q0\right)q0-36\left(-S^2+S^2\,E0^2-L0^2-q0\right)\left(S^2\,E0^2-2\,S\,E0\,L0+L0^2+q0\right)+108\left(-1+E0^2\right)\left(S^2\,E0^2-2\,S\,E0\,L0+L0^2+q0\right)^2\Big)^2\Big)}\Big)^{1/3}\Big)+\frac{1}{2}\sqrt{\Big(\frac{2}{\left(-1+E0^2\right)^2}-\frac{4\left(-S^2+S^2\,E0^2-L0^2-q0\right)}{3\left(-1+E0^2\right)}-\Big(2^{1/3}\Big(\left(-S^2+S^2\,E0^2-L0^2-q0\right)^2-12\,S^2\left(-1+E0^2\right)q0-12\left(S^2\,E0^2-2\,S\,E0\,L0+L0^2+q0\right)\Big)\Big/\Big(3\left(-1+E0^2\right)\Big(2\left(-S^2+S^2\,E0^2-L0^2-q0\right)^3-108\,S^2\,q0+72\,S^2\left(-1+E0^2\right)$$

$$\left(-S^2+S^2\,E0^2-L0^2-q0\right)q0-36\left(-S^2+S^2\,E0^2-L0^2-q0\right)$$
$$\left(S^2\,E0^2-2\,S\,E0\,L0+L0^2+q0\right)+108\left(-1+E0^2\right)\left(S^2\,E0^2-2\,S\,E0\,L0+L0^2+q0\right)^2+$$
$$\sqrt{\left(-4\left(\left(-S^2+S^2\,E0^2-L0^2-q0\right)^2-12\,S^2\left(-1+E0^2\right)q0-\right.\right.}$$
$$\left.12\left(S^2\,E0^2-2\,S\,E0\,L0+L0^2+q0\right)\right)^3+\left(2\left(-S^2+S^2\,E0^2-L0^2-q0\right)^3-$$
$$108\,S^2\,q0+72\,S^2\left(-1+E0^2\right)\left(-S^2+S^2\,E0^2-L0^2-q0\right)q0-$$
$$36\left(-S^2+S^2\,E0^2-L0^2-q0\right)\left(S^2\,E0^2-2\,S\,E0\,L0+L0^2+q0\right)+$$
$$\left.\left.108\left(-1+E0^2\right)\left(S^2\,E0^2-2\,S\,E0\,L0+L0^2+q0\right)^2\right)^2\right)^{1/3}\Bigg)-$$
$$\frac{1}{3\times 2^{1/3}\left(-1+E0^2\right)}\left(2\left(-S^2+S^2\,E0^2-L0^2-q0\right)^3-108\,S^2\,q0+72\,S^2\left(-1+E0^2\right)\right.$$
$$\left(-S^2+S^2\,E0^2-L0^2-q0\right)q0-36\left(-S^2+S^2\,E0^2-L0^2-q0\right)$$
$$\left(S^2\,E0^2-2\,S\,E0\,L0+L0^2+q0\right)+108\left(-1+E0^2\right)\left(S^2\,E0^2-2\,S\,E0\,L0+L0^2+q0\right)^2+$$
$$\sqrt{\left(-4\left(\left(-S^2+S^2\,E0^2-L0^2-q0\right)^2-12\,S^2\left(-1+E0^2\right)q0-12\left(S^2\,E0^2-2\,S\,E0\,L0+\right.\right.\right.}$$
$$\left.\left.L0^2+q0\right)\right)^3+\left(2\left(-S^2+S^2\,E0^2-L0^2-q0\right)^3-108\,S^2\,q0+72\,S^2\left(-1+E0^2\right)\right.$$
$$\left(-S^2+S^2\,E0^2-L0^2-q0\right)q0-36\left(-S^2+S^2\,E0^2-L0^2-q0\right)\left(S^2\,E0^2-2\,S\,E0\right.$$
$$\left.\left.\left.L0+L0^2+q0\right)+108\left(-1+E0^2\right)\left(S^2\,E0^2-2\,S\,E0\,L0+L0^2+q0\right)^2\right)^2\right)^{1/3}-$$
$$\left.\left(-\frac{8}{\left(-1+E0^2\right)^3}+\frac{8\left(-S^2+S^2\,E0^2-L0^2-q0\right)}{\left(-1+E0^2\right)^2}-\frac{16\left(S^2\,E0^2-2\,S\,E0\,L0+L0^2+q0\right)}{-1+E0^2}\right)\right/$$
$$\left(4\sqrt{\left(\frac{1}{\left(-1+E0^2\right)^2}-\frac{2\left(-S^2+S^2\,E0^2-L0^2-q0\right)}{3\left(-1+E0^2\right)}+\left(2^{1/3}\left(\left(-S^2+S^2\,E0^2-L0^2-q0\right)^2-\right.\right.\right.}$$
$$\left.\left.12\,S^2\left(-1+E0^2\right)q0-12\left(S^2\,E0^2-2\,S\,E0\,L0+L0^2+q0\right)\right)\right)\Big/$$
$$\left(3\left(-1+E0^2\right)\left(2\left(-S^2+S^2\,E0^2-L0^2-q0\right)^3-108\,S^2\,q0+72\,S^2\left(-1+E0^2\right)\right.\right.$$
$$\left(-S^2+S^2\,E0^2-L0^2-q0\right)q0-36\left(-S^2+S^2\,E0^2-L0^2-q0\right)\left(S^2\,E0^2-\right.$$
$$\left.2\,S\,E0\,L0+L0^2+q0\right)+108\left(-1+E0^2\right)\left(S^2\,E0^2-2\,S\,E0\,L0+L0^2+q0\right)^2+$$
$$\sqrt{\left(-4\left(\left(-S^2+S^2\,E0^2-L0^2-q0\right)^2-12\,S^2\left(-1+E0^2\right)q0-12\right.\right.}$$
$$\left.\left(S^2\,E0^2-2\,S\,E0\,L0+L0^2+q0\right)\right)^3+\left(2\left(-S^2+S^2\,E0^2-L0^2-q0\right)^3-$$
$$108\,S^2\,q0+72\,S^2\left(-1+E0^2\right)\left(-S^2+S^2\,E0^2-L0^2-q0\right)q0-$$
$$36\left(-S^2+S^2\,E0^2-L0^2-q0\right)\left(S^2\,E0^2-2\,S\,E0\,L0+L0^2+q0\right)+$$
$$\left.\left.\left.108\left(-1+E0^2\right)\left(S^2\,E0^2-2\,S\,E0\,L0+L0^2+q0\right)^2\right)^2\right)^{1/3}\right)+$$
$$\frac{1}{3\times 2^{1/3}\left(-1+E0^2\right)}\left(2\left(-S^2+S^2\,E0^2-L0^2-q0\right)^3-108\,S^2\,q0+72\,S^2\right.$$
$$\left(-1+E0^2\right)\left(-S^2+S^2\,E0^2-L0^2-q0\right)q0-36\left(-S^2+S^2\,E0^2-L0^2-q0\right)\left(S^2\,E0^2-\right.$$

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                2 S E0 L0 + L0^2 + q0) + 108 (-1 + E0^2) (S^2 E0^2 - 2 S E0 L0 + L0^2 + q0)^2 +
          Sqrt[-4 ((-S^2 + S^2 E0^2 - L0^2 - q0)^2 - 12 S^2 (-1 + E0^2) q0 -
                12 (S^2 E0^2 - 2 S E0 L0 + L0^2 + q0))^3 + (2 (-S^2 + S^2 E0^2 - L0^2 - q0)^3 -
              108 S^2 q0 + 72 S^2 (-1 + E0^2) (-S^2 + S^2 E0^2 - L0^2 - q0) q0 -
              36 (-S^2 + S^2 E0^2 - L0^2 - q0) (S^2 E0^2 - 2 S E0 L0 + L0^2 + q0) +
              108 (-1 + E0^2) (S^2 E0^2 - 2 S E0 L0 + L0^2 + q0)^2)^2]]^(1/3)]]]];

radii = {
  Root1 =
   Re[-1/(2 (-1 + Energy[t]^2)) +
     1/2 Sqrt[1/(-1 + Energy[t]^2)^2 - (2 (-S^2 + S^2 Energy[t]^2 - L[t]^2 - q[t]))/(3 (-1 + Energy[t]^2)) +
       (2^(1/3) ((-S^2 + S^2 Energy[t]^2 - L[t]^2 - q[t])^2 - 12 S^2 (-1 + Energy[t]^2) q[t] -
           12 (S^2 Energy[t]^2 - 2 S Energy[t] × L[t] + L[t]^2 + q[t])))/
        (3 (-1 + Energy[t]^2) (2 (-S^2 + S^2 Energy[t]^2 - L[t]^2 - q[t])^3 - 108 S^2 q[t] +
            72 S^2 (-1 + Energy[t]^2) (-S^2 + S^2 Energy[t]^2 - L[t]^2 - q[t]) q[t] -
            36 (-S^2 + S^2 Energy[t]^2 - L[t]^2 - q[t])
             (S^2 Energy[t]^2 - 2 S Energy[t] × L[t] + L[t]^2 + q[t]) + 108
             (-1 + Energy[t]^2) (S^2 Energy[t]^2 - 2 S Energy[t] × L[t] + L[t]^2 + q[t])^2 +
            Sqrt[-4 ((-S^2 + S^2 Energy[t]^2 - L[t]^2 - q[t])^2 - 12 S^2 (-1 + Energy[t]^2) q[t] -
                12 (S^2 Energy[t]^2 - 2 S Energy[t] × L[t] + L[t]^2 + q[t]))^3 +
              (2 (-S^2 + S^2 Energy[t]^2 - L[t]^2 - q[t])^3 - 108 S^2 q[t] +
                72 S^2 (-1 + Energy[t]^2) (-S^2 + S^2 Energy[t]^2 - L[t]^2 - q[t]) q[t] -
                36 (-S^2 + S^2 Energy[t]^2 - L[t]^2 - q[t]) (S^2 Energy[t]^2 -
                  2 S Energy[t] × L[t] + L[t]^2 + q[t]) + 108 (-1 + Energy[t]^2)
                 (S^2 Energy[t]^2 - 2 S Energy[t] × L[t] + L[t]^2 + q[t])^2)^2]]^(1/3)) +
       1/(3 × 2^(1/3) (-1 + Energy[t]^2)) (2 (-S^2 + S^2 Energy[t]^2 - L[t]^2 - q[t])^3 -
          108 S^2 q[t] + 72 S^2 (-1 + Energy[t]^2) (-S^2 + S^2 Energy[t]^2 - L[t]^2 - q[t])
           q[t] - 36 (-S^2 + S^2 Energy[t]^2 - L[t]^2 - q[t])
```

$$\left(S^2\,\text{Energy}[t]^2-2\,S\,\text{Energy}[t]\times L[t]+L[t]^2+q[t]\right)+$$
$$108\left(-1+\text{Energy}[t]^2\right)\left(S^2\,\text{Energy}[t]^2-2\,S\,\text{Energy}[t]\times L[t]+L[t]^2+q[t]\right)^2+$$
$$\sqrt{\left(-4\left(\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)^2-12\,S^2\left(-1+\text{Energy}[t]^2\right)q[t]-\right.\right.}$$
$$\left.12\left(S^2\,\text{Energy}[t]^2-2\,S\,\text{Energy}[t]\times L[t]+L[t]^2+q[t]\right)\right)^3+$$
$$\left(2\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)^3-108\,S^2\,q[t]+72\,S^2\right.$$
$$\left(-1+\text{Energy}[t]^2\right)\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)q[t]-$$
$$36\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)\left(S^2\,\text{Energy}[t]^2-\right.$$
$$2\,S\,\text{Energy}[t]\times L[t]+L[t]^2+q[t]\Big)+108\left(-1+\text{Energy}[t]^2\right)$$
$$\left.\left.\left.\left(S^2\,\text{Energy}[t]^2-2\,S\,\text{Energy}[t]\times L[t]+L[t]^2+q[t]\right)^2\right)^2\right)\Big)^{1/3}\Bigg)+$$

$$\frac{1}{2}\sqrt{\left(\frac{2}{\left(-1+\text{Energy}[t]^2\right)^2}-\frac{4\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)}{3\left(-1+\text{Energy}[t]^2\right)}-\right.}$$
$$\left(2^{1/3}\left(\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)^2-12\,S^2\left(-1+\text{Energy}[t]^2\right)q[t]-\right.\right.$$
$$\left.\left.12\left(S^2\,\text{Energy}[t]^2-2\,S\,\text{Energy}[t]\times L[t]+L[t]^2+q[t]\right)\right)\Big/$$
$$\left(3\left(-1+\text{Energy}[t]^2\right)\left(2\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)^3-108\,S^2\,q[t]+\right.\right.$$
$$72\,S^2\left(-1+\text{Energy}[t]^2\right)\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)q[t]-$$
$$36\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)$$
$$\left(S^2\,\text{Energy}[t]^2-2\,S\,\text{Energy}[t]\times L[t]+L[t]^2+q[t]\right)+108$$
$$\left(-1+\text{Energy}[t]^2\right)\left(S^2\,\text{Energy}[t]^2-2\,S\,\text{Energy}[t]\times L[t]+L[t]^2+q[t]\right)^2+$$
$$\sqrt{\left(-4\left(\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)^2-12\,S^2\left(-1+\text{Energy}[t]^2\right)q[t]-\right.\right.}$$
$$\left.12\left(S^2\,\text{Energy}[t]^2-2\,S\,\text{Energy}[t]\times L[t]+L[t]^2+q[t]\right)\right)^3+$$
$$\left(2\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)^3-108\,S^2\,q[t]+\right.$$
$$72\,S^2\left(-1+\text{Energy}[t]^2\right)\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)q[t]-$$
$$36\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)\left(S^2\,\text{Energy}[t]^2-\right.$$
$$2\,S\,\text{Energy}[t]\times L[t]+L[t]^2+q[t]\Big)+108\left(-1+\text{Energy}[t]^2\right)$$
$$\left.\left.\left.\left(S^2\,\text{Energy}[t]^2-2\,S\,\text{Energy}[t]\times L[t]+L[t]^2+q[t]\right)^2\right)^2\right)\Big)^{1/3}\Bigg)-$$
$$\frac{1}{3\times 2^{1/3}\left(-1+\text{Energy}[t]^2\right)}\left(2\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)^3-\right.$$
$$108\,S^2\,q[t]+72\,S^2\left(-1+\text{Energy}[t]^2\right)\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)$$
$$q[t]-36\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)$$
$$\left(S^2\,\text{Energy}[t]^2-2\,S\,\text{Energy}[t]\times L[t]+L[t]^2+q[t]\right)+$$
$$108\left(-1+\text{Energy}[t]^2\right)\left(S^2\,\text{Energy}[t]^2-2\,S\,\text{Energy}[t]\times L[t]+L[t]^2+q[t]\right)^2+$$
$$\sqrt{\left(-4\left(\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)^2-12\,S^2\left(-1+\text{Energy}[t]^2\right)q[t]-\right.\right.}$$

$$12\left(S^2\,\text{Energy}[t]^2-2\,S\,\text{Energy}[t]\times L[t]+L[t]^2+q[t]\right)\Big)^3+\left(2\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)^3-108\,S^2\,q[t]+72\,S^2\left(-1+\text{Energy}[t]^2\right)\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)q[t]-36\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)\left(S^2\,\text{Energy}[t]^2-2\,S\,\text{Energy}[t]\times L[t]+L[t]^2+q[t]\right)+108\left(-1+\text{Energy}[t]^2\right)\left(S^2\,\text{Energy}[t]^2-2\,S\,\text{Energy}[t]\times L[t]+L[t]^2+q[t]\right)^2\Big)^2\Big)\Big)^{1/3}+\left(-\frac{8}{\left(-1+\text{Energy}[t]^2\right)^3}+\frac{8\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)}{\left(-1+\text{Energy}[t]^2\right)^2}-\left(16\left(S^2\,\text{Energy}[t]^2-2\,S\,\text{Energy}[t]\times L[t]+L[t]^2+q[t]\right)\right)\Big/\left(-1+\text{Energy}[t]^2\right)\Big)\Big/\left(4\sqrt{\left(\frac{1}{\left(-1+\text{Energy}[t]^2\right)^2}-\frac{2\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)}{3\left(-1+\text{Energy}[t]^2\right)}+\left(2^{1/3}\left(\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)^2-12\,S^2\left(-1+\text{Energy}[t]^2\right)q[t]-12\left(S^2\,\text{Energy}[t]^2-2\,S\,\text{Energy}[t]\times L[t]+L[t]^2+q[t]\right)\right)\Big)\Big/\left(3\left(-1+\text{Energy}[t]^2\right)\left(2\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)^3-108\,S^2\,q[t]+72\,S^2\left(-1+\text{Energy}[t]^2\right)\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)q[t]-36\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)\left(S^2\,\text{Energy}[t]^2-2\,S\,\text{Energy}[t]\times L[t]+L[t]^2+q[t]\right)+108\left(-1+\text{Energy}[t]^2\right)\left(S^2\,\text{Energy}[t]^2-2\,S\,\text{Energy}[t]\times L[t]+L[t]^2+q[t]\right)^2+\sqrt{\left(-4\left(\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)^2-12\,S^2\left(-1+\text{Energy}[t]^2\right)q[t]-12\left(S^2\,\text{Energy}[t]^2-2\,S\,\text{Energy}[t]\times L[t]+L[t]^2+q[t]\right)\right)^3+\left(2\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)^3-108\,S^2\,q[t]+72\,S^2\left(-1+\text{Energy}[t]^2\right)\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)q[t]-36\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)\left(S^2\,\text{Energy}[t]^2-2\,S\,\text{Energy}[t]\times L[t]+L[t]^2+q[t]\right)+108\left(-1+\text{Energy}[t]^2\right)\left(S^2\,\text{Energy}[t]^2-2\,S\,\text{Energy}[t]\times L[t]+L[t]^2+q[t]\right)^2\Big)^2\Big)\Big)^{1/3}\Big)+\frac{1}{3\times 2^{1/3}\left(-1+\text{Energy}[t]^2\right)}\Big)\Big)\Big)

$$\left(S^2\,\text{Energy}[t]^2 - 2\,S\,\text{Energy}[t]\times L[t] + L[t]^2 + q[t]\right)^2 +$$
$$\sqrt{\left(-4\left(\left(-S^2 + S^2\,\text{Energy}[t]^2 - L[t]^2 - q[t]\right)^2 - 12\,S^2\left(-1 + \text{Energy}[t]^2\right)\right.\right.}$$
$$q[t] - 12\left(S^2\,\text{Energy}[t]^2 - 2\,S\,\text{Energy}[t]\times L[t] + L[t]^2 +\right.$$
$$\left.\left.q[t]\right)\right)^3 + \left(2\left(-S^2 + S^2\,\text{Energy}[t]^2 - L[t]^2 - q[t]\right)^3 - 108\,S^2\right.$$
$$q[t] + 72\,S^2\left(-1 + \text{Energy}[t]^2\right)\left(-S^2 + S^2\,\text{Energy}[t]^2 -\right.$$
$$\left.L[t]^2 - q[t]\right)q[t] - 36\left(-S^2 + S^2\,\text{Energy}[t]^2 - L[t]^2 - q[t]\right)$$
$$\left(S^2\,\text{Energy}[t]^2 - 2\,S\,\text{Energy}[t]\times L[t] + L[t]^2 + q[t]\right) +$$
$$108\left(-1 + \text{Energy}[t]^2\right)\left(S^2\,\text{Energy}[t]^2 - 2\,S\right.$$
$$\left.\left.\left.\left.\text{Energy}[t]\times L[t] + L[t]^2 + q[t]\right)^2\right)^2\right)\Big)^{1/3}\Big)\Big)\Big)$$

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Root2 =

$$\text{Re}\Big[-\frac{1}{2\left(-1 + \text{Energy}[t]^2\right)} -$$
$$\frac{1}{2}\sqrt{\left(\frac{1}{\left(-1 + \text{Energy}[t]^2\right)^2} - \frac{2\left(-S^2 + S^2\,\text{Energy}[t]^2 - L[t]^2 - q[t]\right)}{3\left(-1 + \text{Energy}[t]^2\right)} +\right.}$$
$$\left(2^{1/3}\left(\left(-S^2 + S^2\,\text{Energy}[t]^2 - L[t]^2 - q[t]\right)^2 - 12\,S^2\left(-1 + \text{Energy}[t]^2\right)q[t] -\right.\right.$$
$$\left.\left.12\left(S^2\,\text{Energy}[t]^2 - 2\,S\,\text{Energy}[t]\times L[t] + L[t]^2 + q[t]\right)\right)\right)\Big/$$
$$\left(3\left(-1 + \text{Energy}[t]^2\right)\left(2\left(-S^2 + S^2\,\text{Energy}[t]^2 - L[t]^2 - q[t]\right)^3 - 108\,S^2\,q[t] +\right.\right.$$
$$72\,S^2\left(-1 + \text{Energy}[t]^2\right)\left(-S^2 + S^2\,\text{Energy}[t]^2 - L[t]^2 - q[t]\right)q[t] -$$
$$36\left(-S^2 + S^2\,\text{Energy}[t]^2 - L[t]^2 - q[t]\right)$$
$$\left(S^2\,\text{Energy}[t]^2 - 2\,S\,\text{Energy}[t]\times L[t] + L[t]^2 + q[t]\right) + 108$$
$$\left(-1 + \text{Energy}[t]^2\right)\left(S^2\,\text{Energy}[t]^2 - 2\,S\,\text{Energy}[t]\times L[t] + L[t]^2 + q[t]\right)^2 +$$
$$\sqrt{\left(-4\left(\left(-S^2 + S^2\,\text{Energy}[t]^2 - L[t]^2 - q[t]\right)^2 - 12\,S^2\left(-1 + \text{Energy}[t]^2\right)q[t] -\right.\right.}$$
$$\left.\left.12\left(S^2\,\text{Energy}[t]^2 - 2\,S\,\text{Energy}[t]\times L[t] + L[t]^2 + q[t]\right)\right)^3 +\right.$$
$$\left(2\left(-S^2 + S^2\,\text{Energy}[t]^2 - L[t]^2 - q[t]\right)^3 - 108\,S^2\,q[t] +\right.$$
$$72\,S^2\left(-1 + \text{Energy}[t]^2\right)\left(-S^2 + S^2\,\text{Energy}[t]^2 - L[t]^2 - q[t]\right)q[t] -$$
$$36\left(-S^2 + S^2\,\text{Energy}[t]^2 - L[t]^2 - q[t]\right)\left(S^2\,\text{Energy}[t]^2 -\right.$$
$$\left.\left.2\,S\,\text{Energy}[t]\times L[t] + L[t]^2 + q[t]\right) + 108\left(-1 + \text{Energy}[t]^2\right)\right.$$
$$\left.\left.\left.\left(S^2\,\text{Energy}[t]^2 - 2\,S\,\text{Energy}[t]\times L[t] + L[t]^2 + q[t]\right)^2\right)^2\right)\Big)^{1/3}\Big) +$$
$$\frac{1}{3\times 2^{1/3}\left(-1 + \text{Energy}[t]^2\right)}\left(2\left(-S^2 + S^2\,\text{Energy}[t]^2 - L[t]^2 - q[t]\right)^3 -\right.$$

$$108\,S^2\,q[t]+72\,S^2\left(-1+\text{Energy}[t]^2\right)\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)q[t]-36\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)\left(S^2\,\text{Energy}[t]^2-2\,S\,\text{Energy}[t]\times L[t]+L[t]^2+q[t]\right)+108\left(-1+\text{Energy}[t]^2\right)\left(S^2\,\text{Energy}[t]^2-2\,S\,\text{Energy}[t]\times L[t]+L[t]^2+q[t]\right)^2+\sqrt{\left(-4\left(\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)^2-12\,S^2\left(-1+\text{Energy}[t]^2\right)q[t]-12\left(S^2\,\text{Energy}[t]^2-2\,S\,\text{Energy}[t]\times L[t]+L[t]^2+q[t]\right)\right)^3+\left(2\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)^3-108\,S^2\,q[t]+72\,S^2\left(-1+\text{Energy}[t]^2\right)\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)q[t]-36\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)\left(S^2\,\text{Energy}[t]^2-2\,S\,\text{Energy}[t]\times L[t]+L[t]^2+q[t]\right)+108\left(-1+\text{Energy}[t]^2\right)\left(S^2\,\text{Energy}[t]^2-2\,S\,\text{Energy}[t]\times L[t]+L[t]^2+q[t]\right)^2\right)^2\right)}^{1/3}+\frac{1}{2}\sqrt{\frac{2}{\left(-1+\text{Energy}[t]^2\right)^2}-\frac{4\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)}{3\left(-1+\text{Energy}[t]^2\right)}-\left(2^{1/3}\left(\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)^2-12\,S^2\left(-1+\text{Energy}[t]^2\right)q[t]-12\left(S^2\,\text{Energy}[t]^2-2\,S\,\text{Energy}[t]\times L[t]+L[t]^2+q[t]\right)\right)\right)\Big/\left(3\left(-1+\text{Energy}[t]^2\right)\left(2\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)^3-108\,S^2\,q[t]+72\,S^2\left(-1+\text{Energy}[t]^2\right)\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)q[t]-36\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)\left(S^2\,\text{Energy}[t]^2-2\,S\,\text{Energy}[t]\times L[t]+L[t]^2+q[t]\right)+108\left(-1+\text{Energy}[t]^2\right)\left(S^2\,\text{Energy}[t]^2-2\,S\,\text{Energy}[t]\times L[t]+L[t]^2+q[t]\right)^2+\sqrt{\left(-4\left(\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)^2-12\,S^2\left(-1+\text{Energy}[t]^2\right)q[t]-12\left(S^2\,\text{Energy}[t]^2-2\,S\,\text{Energy}[t]\times L[t]+L[t]^2+q[t]\right)\right)^3+\left(2\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)^3-108\,S^2\,q[t]+72\,S^2\left(-1+\text{Energy}[t]^2\right)\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)q[t]-36\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)\left(S^2\,\text{Energy}[t]^2-2\,S\,\text{Energy}[t]\times L[t]+L[t]^2+q[t]\right)+108\left(-1+\text{Energy}[t]^2\right)\left(S^2\,\text{Energy}[t]^2-2\,S\,\text{Energy}[t]\times L[t]+L[t]^2+q[t]\right)^2\right)^2\right)}^{1/3}\right)-\frac{1}{3\times 2^{1/3}\left(-1+\text{Energy}[t]^2\right)}\left(2\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)^3-108\,S^2\,q[t]+72\,S^2\left(-1+\text{Energy}[t]^2\right)\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)q[t]-36\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)\left(S^2\,\text{Energy}[t]^2-2\,S\,\text{Energy}[t]\times L[t]+L[t]^2+q[t]\right)+$$

$$108\left(-1+\text{Energy}[t]^2\right)\left(S^2\,\text{Energy}[t]^2-2\,S\,\text{Energy}[t]\times L[t]+L[t]^2+q[t]\right)^2+\sqrt{\left(-4\left(\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)^2-12\,S^2\left(-1+\text{Energy}[t]^2\right)q[t]-12\left(S^2\,\text{Energy}[t]^2-2\,S\,\text{Energy}[t]\times L[t]+L[t]^2+q[t]\right)\right)^3+\left(2\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)^3-108\,S^2\,q[t]+72\,S^2\left(-1+\text{Energy}[t]^2\right)\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)q[t]-36\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)\left(S^2\,\text{Energy}[t]^2-2\,S\,\text{Energy}[t]\times L[t]+L[t]^2+q[t]\right)+108\left(-1+\text{Energy}[t]^2\right)\left(S^2\,\text{Energy}[t]^2-2\,S\,\text{Energy}[t]\times L[t]+L[t]^2+q[t]\right)^2\right)^2\right)}^{1/3}-\left(-\frac{8}{\left(-1+\text{Energy}[t]^2\right)^3}+\frac{8\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)}{\left(-1+\text{Energy}[t]^2\right)^2}-\left(16\left(S^2\,\text{Energy}[t]^2-2\,S\,\text{Energy}[t]\times L[t]+L[t]^2+q[t]\right)\right)/\left(-1+\text{Energy}[t]^2\right)\right)\Bigg/\left(4\sqrt{\left(\frac{1}{\left(-1+\text{Energy}[t]^2\right)^2}-\frac{2\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)}{3\left(-1+\text{Energy}[t]^2\right)}+\left(2^{1/3}\left(\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)^2-12\,S^2\left(-1+\text{Energy}[t]^2\right)q[t]-12\left(S^2\,\text{Energy}[t]^2-2\,S\,\text{Energy}[t]\times L[t]+L[t]^2+q[t]\right)\right)\right)\Bigg/\left(3\left(-1+\text{Energy}[t]^2\right)\left(2\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)^3-108\,S^2\,q[t]+72\,S^2\left(-1+\text{Energy}[t]^2\right)\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)q[t]-36\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)\left(S^2\,\text{Energy}[t]^2-2\,S\,\text{Energy}[t]\times L[t]+L[t]^2+q[t]\right)+108\left(-1+\text{Energy}[t]^2\right)\left(S^2\,\text{Energy}[t]^2-2\,S\,\text{Energy}[t]\times L[t]+L[t]^2+q[t]\right)^2+\sqrt{\left(-4\left(\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)^2-12\,S^2\left(-1+\text{Energy}[t]^2\right)q[t]-12\left(S^2\,\text{Energy}[t]^2-2\,S\,\text{Energy}[t]\times L[t]+L[t]^2+q[t]\right)\right)^3+\left(2\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)^3-108\,S^2\,q[t]+72\,S^2\left(-1+\text{Energy}[t]^2\right)\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)q[t]-36\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)\left(S^2\,\text{Energy}[t]^2-2\,S\,\text{Energy}[t]\times L[t]+L[t]^2+q[t]\right)+108\left(-1+\text{Energy}[t]^2\right)\left(S^2\,\text{Energy}[t]^2-2\,S\,\text{Energy}[t]\times L[t]+L[t]^2+q[t]\right)^2\right)^2\right)}^{1/3}\right)+\frac{1}{3\times 2^{1/3}\left(-1+\text{Energy}[t]^2\right)}\right)\left(2\left(-S^2+S^2\,\text{Energy}[t]^2-L[t]^2-q[t]\right)^3-108\,S^2\,q[t]+72\,S^2\right)$$

```
        (-1 + Energy[t]^2) (-S^2 + S^2 Energy[t]^2 - L[t]^2 - q[t]) q[t] -
      36 (-S^2 + S^2 Energy[t]^2 - L[t]^2 - q[t]) (S^2 Energy[t]^2 -
         2 S Energy[t] × L[t] + L[t]^2 + q[t]) + 108 (-1 + Energy[t]^2)
        (S^2 Energy[t]^2 - 2 S Energy[t] × L[t] + L[t]^2 + q[t])^2 +
      √(-4 ((-S^2 + S^2 Energy[t]^2 - L[t]^2 - q[t])^2 - 12 S^2 (-1 + Energy[t]^2)
              q[t] - 12 (S^2 Energy[t]^2 - 2 S Energy[t] × L[t] + L[t]^2 +
                 q[t]))^3 + (2 (-S^2 + S^2 Energy[t]^2 - L[t]^2 - q[t])^3 - 108 S^2
            q[t] + 72 S^2 (-1 + Energy[t]^2) (-S^2 + S^2 Energy[t]^2 -
              L[t]^2 - q[t]) q[t] - 36 (-S^2 + S^2 Energy[t]^2 - L[t]^2 - q[t])
            (S^2 Energy[t]^2 - 2 S Energy[t] × L[t] + L[t]^2 + q[t]) +
          108 (-1 + Energy[t]^2) (S^2 Energy[t]^2 - 2 S
               Energy[t] × L[t] + L[t]^2 + q[t])^2)^2)^(1/3)))))
  ]};

(*Assigning the roots to Ra or Rp*)
If[Root20 < Root10,
 (Ra[t_] = Root1; Rp[t_] = Root2),
 (Rp[t_] = Root1; Ra[t_] = Root2)
];

(*Defining the semi-latus rectum and eccentricity in terms of Ra and Rp.*)
p[t_] = 2 Rp[t] × Ra[t] / (Rp[t] + Ra[t]);
e[t_] = (Ra[t] - Rp[t]) / (Ra[t] + Rp[t]);
(*Defining the inclination angle.*)
ι[t_] = ArcTan[√(q[t] / L[t]^2)];
(*Time-evolution equations for energy, angular momentum,
and Carter constant from (44-45) of Gair & Glampedakis (0510129)*)
dE = (-32/5) * ν * (1/p[t])^5 (1 - e[t]^2)^(3/2)
  ((1 + 73/24 e[t]^2 + 37/96 e[t]^4) - a * (1/p[t])^(3/2) (73/12 + 823/24 e[t]^2 + 949/32 e[t]^4 + 491/192 e[t]^6)
     Cos[ι[t]] - (1/p[t]) (1247/336 + 9181/672 e[t]^2) + Pi (1/p[t])^(3/2) (4 + 1375/48 e[t]^2) - (1/p[t])^2
```

```
      (44711/9072 + 172157/2592 e[t]^2) + a^2 (1/p[t])^2 (33/16 + 359/32 e[t]^2) - 527/96 a^2 (1/p[t])^2 Sin[ι[t]]^2);
dL = -32/5 ν (1/p[t])^(7/2) (1 - e[t]^2)^(3/2) ((1 + 7/8 e[t]^2) Cos[ι[t]] +
     a (1/p[t])^(3/2) ((61/24 + 63/8 e[t]^2 + 95/64 e[t]^4) - Cos[ι[t]]^2 (61/8 + 91/4 e[t]^2 + 461/64 e[t]^4)) -
     (1/p[t]) (1247/336 + 425/336 e[t]^2) Cos[ι[t]] + Pi (1/p[t])^(3/2) (4 + 97/8 e[t]^2) Cos[ι[t]] -
     (1/p[t])^2 (44711/9072 + 302893/6048 e[t]^2) Cos[ι[t]] +
     a^2 (1/p[t])^2 Cos[ι[t]] ((33/16 + 95/16 e[t]^2) - 45/8 Sin[ι[t]]^2));
(*dQ/dt is (56)-0510129-  *)
dQ = -64/5 ν (1/p[t])^(7/2) Sqrt[q[t]] Sin[ι[t]] (1 - e[t]^2)^(3/2)
   ((1 + 7/8 e[t]^2) - a (1/p[t])^(3/2) Cos[ι[t]] (61/8 + 91/4 e[t]^2 + 461/64 e[t]^4) -
     (1/p[t]) (1247/336 + 425/336 e[t]^2) + Pi (1/p[t])^(3/2) (4 + 97/8 e[t]^2) -
     (1/p[t])^2 (44711/9072 + 302893/6048 e[t]^2) + a^2 (1/p[t])^2 ((33/16 + 95/16 e[t]^2) - 45/8 Sin[ι[t]]^2));

(*dQ/dt is (61)-0510129-*)
dQ2 = -64/5 ν (1/p[t])^(7/2) Sqrt[q[t]] Sin[ι[t]] (1 - e[t]^2)^(3/2)
   ((1 + 7/8 e[t]^2) - a (1/p[t])^(3/2) Cos[ι[t]] (61/8 + 91/4 e[t]^2 + 461/64 e[t]^4) -
     (1/p[t]) (1247/336 + 425/336 e[t]^2) + Pi (1/p[t])^(3/2) (4 + 97/8 e[t]^2) -
     (1/p[t])^2 (44711/9072 + 302893/6048 e[t]^2) + a^2 (1/p[t])^2 (211/96 + 847/96 e[t]^2));

(*Redefining some orbital constants.*)
zm = Cos[Pi/2 - ι[t]]^2;
Δ = r^2 - 2 r + a^2;
Δ2 = Δ /. r → p[t]/(1 + e[t] * Cos[Psi[t]]);
rpsi = p[t]/(1 + e[t] * Cos[Psi[t]]);
```

```
(*Defining ODEs of particle trajectory-1109.0572-*)
zChi = zm * (Cos[Chi[t]])^2;
zp = q[t] / (a^2 (1 - (Energy[t])^2) zm);
dChi = Sqrt[a^2 (1 - (Energy[t])^2) (zp - zChi)] / (PSI + a^2 (Energy[t]) zChi);
PSI = ((rpsi^2 + a^2)^2 / Δ2 - a^2) (Energy[t]) - 2 a rpsi L[t] / Δ2;
r3 = (1 / (1 - (Energy[t])^2) - (Ra[t] + Rp[t]) / 2) +
   Sqrt[((1 / (1 - (Energy[t])^2) - (Ra[t] + Rp[t]) / 2)^2 - a^2 q[t] / (1 - (Energy[t])^2) * 1 / (Ra[t] × Rp[t]))];
r4 = (1 / (1 - (Energy[t])^2) - (Ra[t] + Rp[t]) / 2) -
   Sqrt[((1 / (1 - (Energy[t])^2) - (Ra[t] + Rp[t]) / 2)^2 - a^2 q[t] / (1 - (Energy[t])^2) * 1 / (Ra[t] × Rp[t]))];
p3 = r3 (1 - e[t]);
p4 = r4 (1 + e[t]);
dpsi =
  (Sqrt[1 - (Energy[t])^2] Sqrt((p[t] (1 - e[t]) - p3 (1 + e[t] Cos[Psi[t]])) * (p[t] (1 + e[t]) -
          p4 (1 + e[t] Cos[Psi[t]])))) / ((1 - e[t]^2) (PSI + a^2 (Energy[t]) zChi));
dPhi = 1 / (PSI + a^2 (Energy[t]) zChi) (2 a rpsi (Energy[t]) / Δ2 + (1 / (1 - zChi) - a^2 / Δ2) * L[t]);

sol3 = NDSolve[{Energy'[t] == dE, L'[t] == dL, q'[t] == dQ2,
    Psi'[t] == dpsi, Chi'[t] == dChi, Phi'[t] == dPhi, Energy[0] == Epro,
    L[0] == Lpro, q[0] == Q, Psi[0] == 0, Chi[0] == i * (Pi / 180), Phi[0] == 0},
  {Energy[t], L[t], q[t], Psi[t], Chi[t], Phi[t]}, {t, 0, 10^8},
  Method → {"EquationSimplification" → "Residual"}(*,MaxSteps→Infinity*)];

(*Assigning numerical solver outputs to time-dependent functions.*)
E1[t_] = Energy[t] /. sol3[[1]][[1]];
L1[t_] = L[t] /. sol3[[1]][[2]];
Q1[t_] = q[t] /. sol3[[1]][[3]];
X1[t_] = Psi[t] /. sol3[[1]][[4]](*Eq for Psi(t)*);
Y1[t_] = Chi[t] /. sol3[[1]][[5]] (*Eq for Chi(t)*);
Z1[t_] = Phi[t] /. sol3[[1]][[6]] (*Eq for Phi(t)*);
tf = E1["Domain"][[1]][[2]];
```

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```
Energy[t_] := E1[t];
L[t_] := L1[t];
q[t_] := Q1[t];

(*Eqs of motion in Cartesian Coordinates*)
(X[t_] =
  (p[t] / (1 + e[t] * Cos[X1[t]])) * Sin[ArcCos[(zm^0.5) * Cos[Y1[t]]]] * Cos[Z1[t]]);
(Y[t_] = (p[t] / (1 + e[t] * Cos[X1[t]])) *
   Sin[ArcCos[(zm^0.5) * Cos[Y1[t]]]] * Sin[Z1[t]]);
(Z[t_] = (p[t] / (1 + e[t] * Cos[X1[t]])) * Cos[ArcCos[(zm^0.5) * Cos[Y1[t]]]]);

(*Calculating the Kerr LSO
 (via Stein & Warburton-1912.07609- to halt GW plotting at LSO*)
(*Calculates the last stable orbit in Kerr spacetime. As spin→0,
r→6M+2e. As spin→0 and ecc→0, r→6M*)
s = Sin[i * Pi/180];
S0 = a^12 (-1 + e)^4 (1 + e)^8 s^4;
S1 = -4 a^10 (-3 + e) (-1 + e)^3 (1 + e)^7 s^4;
S2 = 2 a^8 (-1 + e)^2 (1 + e)^6 (2 (-3 + e)^2 + a^2 (3 + e (-2 + 3 e))) s^4;
S3 = -4 a^8 (-1 + e) (1 + e)^5 (-7 + e (7 + e (-13 + 5 e))) s^4;
S4 = a^6 (1 + e)^4 s^2 (16 (-3 + e) (-1 + e)^2 (1 + e) s^2 +
     a^2 (2 (-3 + e) (-1 + e)^2 (1 + e) + (21 + e (-28 + e (22 + e (-12 + 13 e)))) s^2));
S5 = -8 a^6 (-1 + e) (1 + e)^3 s^2 (3 + e - e^2 + e^3 + 2 (-3 - e + 2 e^3) s^2);
S6 = 4 a^4 (1 + e)^2 s^2 (2 (-1 + e^2) (7 + e^2 + 2 (-9 + e^2) s^2) +
     a^2 (-4 + 9 s^2 + e^2 (-2 + 2 (-2 + e) e + (8 + e (4 + 3 e)) s^2)));
S7 = 8 a^4 (1 + e)^2 s^2 (15 - 5 e + e^2 - 3 e^3 - 2 (9 + e (-3 + e + e^2)) s^2);
S8 = 16 a^2 (-1 + e) (1 + e)^2 (3 + e) s^2 +
   a^4 ((-3 + e)^2 (1 + e)^2 + 2 (1 + e) (-15 + e (-5 + e (-1 + 5 e))) s^2 + 4 (3 + e (2 + e))^2 s^4);
S9 = -4 a^2 (7 + e (7 + e + e^2 + 4 e (3 + e) s^2));
S10 = 4 (3 + e)^2 + 2 a^2 (-3 - 2 e + e^2 + 2 (3 + e (2 + e)) s^2);
S11 = -4 (3 + e);
S12 = 1;
sep_poly = p /. {ToRules[Roots[Sum_{n=0}^{12} Sn p^n == 0, p]]};
sep_pol = Quiet[Select[
    p /. {ToRules[Roots[p^5 (-6 - 2 e + p) + a^2 p^3 (-4 (-1 + e) (1 + e)^2 + (3 + e (2 + 3 e)) p) -
          a^4 (1 + e)^2 p (6 + 2 e^3 + 2 e (-1 + p) - 3 p - 3 e^2 (2 + p)) + a^6 (-1 + e)^2 (1 + e)^4 ==
        0, p]]}, 1 + Sqrt[3] + Sqrt[3 + 2 Sqrt[3]] <= # <= 8 &]][[1]];
sep_equat =
 Quiet[Select[p /. {ToRules[Roots[a^4 (-3 - 2 e + e^2)^2 + p^2 (-6 - 2 e + p)^2 - 2 a^2 (1 + e) p (14 +
```

```
                2 e^2 + 3 p - e p) == 0, p]]}, 1 + e ≤ # ≤ 6 + 2 e &]][[1]];
    LSO = Quiet[Select[sep_poly, sep_equat ≤ # ≤ sep_pol &]][[1]];

    If[(p[tf] / (1 + e[tf] * Cos[X1[tf]])) < LSO,
     Tf = Length[Select[(p[Range[0, tf]] / (1 + e[Range[0, tf]] * Cos[X1[Range[0, tf]]])),
        # > LSO &]], Tf = tf - 2];

  };
```

In[162]:=

```
KerrRad[a = 0.9, p = 40, e = 0.9, i = 45, ν = 1, Θ = Pi, ϕ = 0, M = 5]
```

NDSolve: For the method IDA, only machine real code is available. Unable to continue with complex values or beyond floating-point exceptions.

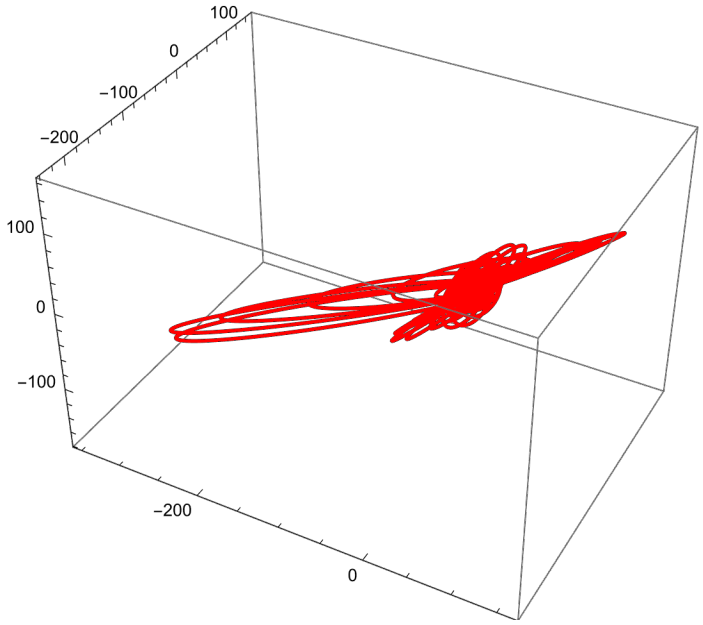
Out[162]=

```
$Aborted
```

In[163]:=

```
(*Running the function with initical parameters of choice.*)
ParametricPlot3D[
 { (p[t] / (1 + e[t] * Cos[X1[t]])) * Sin[ArcCos[(zm^0.5) * Cos[Y1[t]]]] * Cos[Z1[t]],
  (p[t] / (1 + e[t] * Cos[X1[t]])) * Sin[ArcCos[(zm^0.5) * Cos[Y1[t]]]] * Sin[Z1[t]],
  (p[t] / (1 + e[t] * Cos[X1[t]])) * Cos[ArcCos[(zm^0.5) * Cos[Y1[t]]]]},
 {t, 0, tf}, PlotStyle → Red, PlotPoints → 500]
```

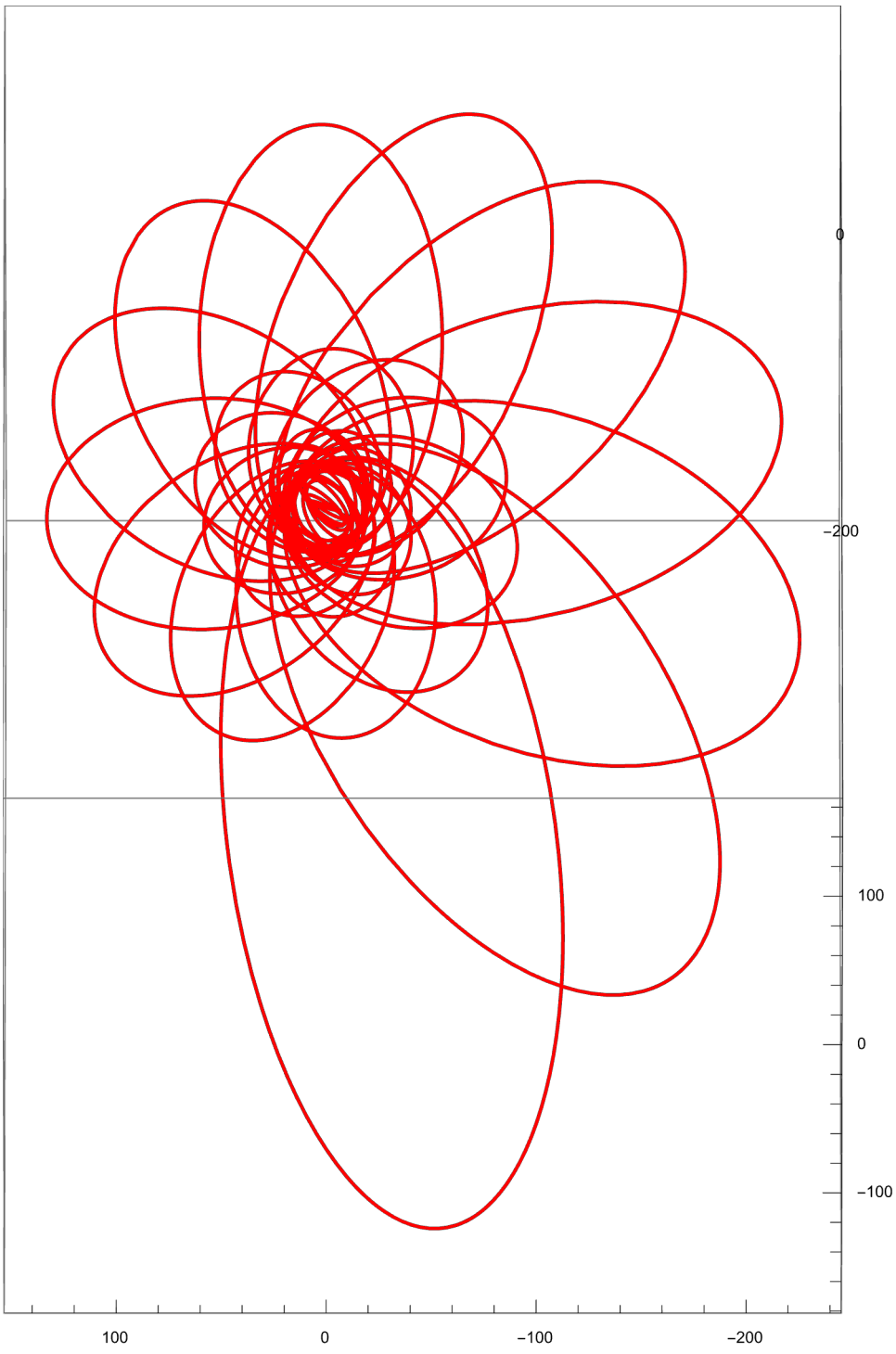
Out[163]=



In[164]:=

```
Show[%, ViewPoint → {-100, 0, 100}, ImageSize → 500]
```

Out[164]=



```
(*Animation Block*)
(*Defining an array of zeros for later population of movie frames.*)
dt = 100;
tRange = Range[1, tf, dt];
movievector = ConstantArray[0, Length[tRange]];

For[k = 1, k ≤ Length[tRange], k++,
 tk = tRange[[k]];
 movievector[[k]] =
  Show[{
    Graphics3D[{Black, Sphere[{0, 0, 0}, 1 + √(1^2 - a^2)]}],
    ParametricPlot3D[{X[t], Y[t], Z[t]}, {t, tk - 10000, tk}, PlotStyle → Thickness[0.01],
     ColorFunction → Function[{x, y, z, t}, Opacity[t^1, Red]]]},
   PlotRange → {{-375, 200}, {-250, 150}, {-200, 150}}, Boxed → False,
   PlotPoints → 500, ViewPoint → {-100, 0, 100}, ImageSize → 500]
]
Export["KerrEccNoAxes.mp4", movievector]
```